Register No															

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
21CSE583T – NATURAL LANGUAGE PROCESSING AND ITS APPLICATIONS
CYCLE TEST – II

Program Offered: M. Tech
Max. Marks: 40

Year / Semester: I / I
Duration: 1.5 Hrs
Date of Exam:

Part - A (2 * 20 = 40 Marks)

Answer any two Questions

Model Question Paper

1)	You are working as a Data Scientist for a Natural Language Processing (NLP) project aimed at developing an intelligent chatbot for customer support. To enhance the chatbot's ability to understand and generate human-like responses, you are considering multiple approaches, including N-Gram models, Skip-Gram, and deep learning-based sequence modeling. a) Explain how an N-Gram Model and Skip-Gram Model can be used to predict the next word in a sentence. What are the limitations of N-Gram models in handling long-term dependencies? (4 marks) b) In the context of document similarity, explain how N-Gram-based similarity and deep learning models differ in computing similarity between two documents. (3 marks) c) Sequence modeling plays a crucial role in NLP tasks. Why is recurrence necessary in sequence modeling? Explain how Recurrent Neural Networks (RNNs) address sequential dependencies in text processing. (3 marks) You are a Machine Learning Engineer working on a speech-to-text application that transcribes spoken language into written text. To improve the accuracy of the transcription, you are considering different Recurrent Neural Network (RNN) architectures, including LSTM, Bi-Directional LSTM, and Multilayer LSTMs. Case Study: The model needs to handle long-term dependencies in speech.

	Training the model requires optimizing weights efficiently over sequential data. You are evaluating different RNN-based architectures to improve performance. d) Explain the role of Backpropagation Through Time (BPTT) in training RNNs. Why does it suffer from vanishing and exploding gradient problems? (3 marks) e) LSTMs use gates to manage information flow. Describe the Forget Gate, Input Gate, and Output Gate in an LSTM cell and their importance in long-term dependency handling. (3 marks) f) How does a Bi-Directional LSTM differ from a standard LSTM? When would using a Multilayer LSTM be beneficial in an NLP task like speech-to-text? (4 marks)
2)	You are leading a research team developing an AI-powered translation system that translates speech from one language to another in real-time. To achieve high accuracy, your team is exploring Attention Mechanisms, Large Language Models (LLMs), and Encoder-Decoder Architectures. Case Study: The system needs to handle long sentences while maintaining context. Traditional sequence-to-sequence models struggle with long-term dependencies. You are evaluating transformer-based architectures to improve translation quality. a) Explain the role of the Attention Mechanism in overcoming the limitations of traditional sequence-to-sequence models. How does it improve context retention? (3 marks) b) Describe the Encoder-Decoder Architecture and its importance in machine translation. How does the addition of the Attention Mechanism enhance the performance of this architecture? (4 marks) c) Large Language Models (LLMs) like GPT and BERT have revolutionized NLP tasks. How do LLMs differ from traditional encoder-decoder models in handling language understanding and generation? (3 marks) You are developing an AI-powered chatbot for a multilingual e-commerce platform that serves users in multiple Indian languages. To ensure smooth interactions, your team is exploring Transformer-based models like BERT, GPT, and XLNet.

	d) Explain how the Transformer architecture differs from traditional RNN-based models in handling sequential text. (3 marks) e) Compare BERT, XLNet, and GPT in terms of their pretraining objectives and how they impact downstream NLP tasks. (4 marks) f) Given that your chatbot needs context-aware responses in multiple Indian languages, which of these models would you prioritize, and why? (3 marks)
3)	A government initiative in India is developing an automated legal document translation system that translates and summarizes court judgments across multiple regional languages. You are evaluating models like MuRIL BERT, mBERT, and mBART for this task. a) What are the key challenges in developing Large Language Models (LLMs) for Indian languages, and how do multilingual models address these challenges? (3 marks) b) Compare mBERT, mBART, and MuRIL BERT in terms of their strengths and limitations for multilingual text processing. (4 marks) c) If the goal is to translate and summarize legal texts with high accuracy across multiple Indian languages, which model would you recommend and why? (3 marks) Your team is building a sentiment analysis model for a social media platform that analyzes user comments to detect positive, negative, and neutral sentiments. You are comparing Multilayer LSTM and Bi-Directional LSTM for better performance. a) What are the advantages of using an LSTM over a simple RNN for sentiment analysis? (3 marks) b) How does a Bi-Directional LSTM differ from a standard LSTM, and why might it be useful for sentiment analysis? (4 marks) c) If your sentiment analysis model needs to capture both past and future context in user comments, would you choose a Multilayer LSTM or a Bi-Directional LSTM? Explain your reasoning. (3 marks)